

Graph the functions, draw a sample of a cross section of an arbitrary shell, and then compute. (Hint: You will sometimes need to use multiple integrals.)

1. Use the shell method to find the volume of the solid obtained by revolving the region enclosed by $y = \sqrt{x+1}$, $y = 1$, $x = 0$, and $x = 3$ about the y -axis.
2. Use the shell method to find the volume of the solid obtained by revolving the regions enclosed by $y = x^2$, $y = 2 - x$, $x = 0$, and $x = 2$ about the y -axis.

3. Use the shell method to find the volume of the solid obtained by revolving the region enclosed by $y = 1 + \sqrt{x}$, $y = 1 - \sqrt{x}$, $x = 0$, and $x = 1$ about the x -axis.

4. Use the shell method to find the volume of the solid obtained by revolving the regions enclosed by $y = x + 1$, $y = 2 - x$, and $x = 0$ and $x = 2$ about the line $y = -1$.

5. Now use a combination of disks and shells. Find the volume of the solid obtained by revolving the regions enclosed by $y = x + 1$, $y = 2 - x$, and $x = 0$ and $x = 2$ about the y -axis. A cross-section of this solid should be a wonky bow tie looking thing.

Use disks for the 'left' piece (inner solid) and shells for the 'right' piece (outer solid) or vice versa.